

CHAPTER 6.2

VOLUMES BY SLICING; DISKS AND WASHERS

Volumes by slicing

6.2.1 Volume Problem

Let S be a solid that extends along the x -axis and is bounded on the left and right, respectively, by the planes that are perpendicular to the x -axis at $x = a$ and $x = b$ (Figure 6.2.5). Find the volume V of the solid, assuming that its cross-sectional area $A(x)$ is known at each x in the interval $[a, b]$.

6.2.2 Volume Formula

Let S be a solid bounded by two parallel planes perpendicular to the x -axis at $x = a$ and $x = b$. If, for each x in $[a, b]$, the cross-sectional area of S perpendicular to the x -axis is $A(x)$, then the volume of the solid is

$$V = \int_a^b A(x) dx \quad (1)$$

provided $A(x)$ is integrable.

6.2.3 Volume Formula

Let S be a solid bounded by two parallel planes perpendicular to the y -axis at $y = c$ and $y = d$. If, for each y in $[c, d]$, the cross-sectional area of S perpendicular to the y -axis is $A(y)$, then the volume of the solid is

$$V = \int_c^d A(y) dy \quad (2)$$

provided $A(y)$ is integrable.

The above formulas state that the volume of a solid can be obtained by integrating the cross-sectional area from one end of the solid to the other.

Example 1

Volume of a right pyramid

Solids of Revolution

A solid of revolution is a solid that is generated by revolving a plane region about a line that lies in the same plane as the region; the line is called the *axis of revolution*. Many familiar solids are of this type, for example, right circular cylinder, solid sphere, solid cone, hollowed right circular cylinder (Figure 6.2.8).

VOLUMES BY DISKS PERPENDICULAR TO THE x -AXIS

6.2.4 Problem Let f be continuous and nonnegative on $[a, b]$, and let R be the region that is bounded above by $y = f(x)$, below by the x -axis, and on the sides by the lines $x = a$ and $x = b$ (Figure 6.2.9a). Find the volume of the solid of revolution that is generated by revolving the region R about the x -axis.

Volume Formula

We can solve this problem by slicing. For this purpose, observe that the cross section of the solid taken perpendicular to the x -axis at the point x is a circular disk of radius $f(x)$ (Figure 6.2.9b). The area of this region is $A(x) = \pi\{f(x)\}^2$.

Thus, the volume of the solid is

$$V = \int_a^b \pi\{f(x)\}^2 dx \quad (3)$$

Example 2

Find the volume of the solid that is obtained when the region under the curve $y = \sqrt{x}$ over the interval $[1, 4]$ is revolved about the x -axis (Figure 6.2.10).

Solution: Volume of the solid, $V = \int_a^b \pi\{f(x)\}^2 dx = \pi \int_1^4 x dx = \frac{15\pi}{2}$

Example 3

Derive the formula for the volume of a sphere of radius r .

Solution: Volume of the sphere, $V = \int_{-r}^r \pi(r^2 - x^2) dx = \frac{4}{3}\pi r^3$

VOLUMES BY WASHERS PERPENDICULAR TO THE x -AXIS

6.2.5 Problem Let f and g be continuous and nonnegative on $[a, b]$, and suppose that $f(x) \geq g(x)$ for all x in the interval $[a, b]$. Let R be the region that is bounded above by $y = f(x)$, below by $y = g(x)$, and on the sides by the lines $x = a$ and $x = b$ (Figure 6.2.12a). Find the volume of the solid of revolution that is generated by revolving the region R about the x -axis (Figure 6.2.12b).

Volume Formula

We can solve this problem by slicing. For this purpose, observe that the cross section of the solid taken perpendicular to the x -axis at the point x is the annular or “washer-shaped” region with inner radius $g(x)$ and outer radius $f(x)$ (Figure 6.2.12b); its area is $A(x) = \pi\{f(x)\}^2 - \pi\{g(x)\}^2$.

Thus, the volume of the solid is

$$V = \int_a^b \pi[\{f(x)\}^2 - \{g(x)\}^2] dx \quad (4)$$

Example 4

Find the volume of the solid generated when the region between the graphs of the equations $f(x) = \frac{1}{2} + x^2$ and $g(x) = x$ over the interval $[0, 2]$, is revolved about the x -axis.

Solution: Volume, $V = \int_a^b \pi[\{f(x)\}^2 - \{g(x)\}^2] dx = \int_0^2 \pi\left(\left(\frac{1}{2} + x^2\right)^2 - x^2\right) dx = \frac{69\pi}{10}$

VOLUMES BY DISKS AND WASHERS PERPENDICULAR TO THE y -AXIS

Volume formula by Disks:

$$V = \int_c^d \pi\{u(y)\}^2 dy \quad (5)$$

Volume formula by Washers:

$$V = \int_c^d \pi[\{w(y)\}^2 - \{v(y)\}^2] dy \quad (6)$$

Example 5

Find the volume of the solid generated when the region enclosed by $y = \sqrt{x}$, $y = 2$, and $x = 0$ is revolved about the y -axis.

Solution: First sketch the region and the solid (Figure 6.2.16). The cross sections taken perpendicular to the y -axis are disks, so we will apply (5). But first we must rewrite $y = \sqrt{x}$ as $x = y^2$. Thus, from (5) with $u(y) = y^2$. The volume is

$$V = \int_c^d \pi \{u(y)\}^2 dy = \pi \int_0^2 y^4 dy = \frac{32\pi}{5}$$

OTHER AXES OF REVOLUTION

It is possible to use the method of disks and the method of washers to find the volume of a solid of revolution whose axis of revolution is a line other than one of the coordinate axes. Instead of developing a new formula for each situation, we will appeal to Formulas (1) and (2) and integrate an appropriate cross-sectional area to find the volume.

Example 6

Find the volume of the solid generated when the region under the curve $y = x^2$ over the interval $[0, 2]$ is rotated about the line $y = -1$.

Solution. First sketch the region and the axis of revolution; then imagine revolving the region about the axis (Figure 6.2.17). At each x in the interval $0 \leq x \leq 2$, the cross section of the solid perpendicular to the axis $y = -1$ is a washer with outer radius $y = x^2 + 1$ and inner radius 1. Since the area of this washer is

$$A(x) = \pi[(x^2 + 1)^2 - 1^2] = \pi(x^4 + 2x^2).$$

The volume is

$$V = \int_0^2 A(x) dx = \pi \int_0^2 (x^4 + x^2) dx = \frac{176\pi}{15}$$

EXERCISE SET 6.2

Find the volume of the solid that results when the region enclosed by the given curves is revolved about the x -axis (11-18):

11. $y = \sqrt{25 - x^2}$, $y = 3$

Since the line $y = 3$ intersects the graph $y = \sqrt{25 - x^2}$, we have

$$3 = \sqrt{25 - x^2} \Rightarrow 9 = 25 - x^2 \Rightarrow x = -4, 4$$

Thus, the volume of the solid is

$$\begin{aligned} V &= \pi \int_a^b [\{f(x)\}^2 - \{g(x)\}^2] dx = \pi \int_{-4}^4 [(25 - x^2) - 9] dx \\ &= 2\pi \int_0^4 (16 - x^2) dx = 2\pi \left(16x - \frac{x^3}{3} \right) \Big|_0^4 = 2\pi \left(64 - \frac{64}{3} \right) = \frac{256\pi}{3} \end{aligned}$$

12. $y = 9 - x^2$, $y = 0$

The volume of the solid is

$$\begin{aligned} V &= \pi \int_{-3}^3 \{f(x)\}^2 dx = \pi \int_{-3}^3 (9 - x^2)^2 dx = 2\pi \int_0^3 (81 - 18x^2 + x^4) dx \\ &= 2\pi \left(81x - 6x^3 + \frac{x^5}{5} \right) \Big|_0^3 = 2\pi \left(324 - 384 + \frac{64 \times 16}{5} \right) = \frac{1296\pi}{5} \end{aligned}$$

13. $x = \sqrt{y}$, $x = y/4$

If the given graphs intersect, then

$$\sqrt{y} = y/4 \Rightarrow y = \frac{y^2}{16} \Rightarrow y = 0, 16$$

At $y = 0$, we have $x = 0$ and at $y = 16$, we have $x = 4$

Hence $(0, 0)$ and $(4, 16)$ are the points of intersection of the line $y = 4x$ with the parabola $y = x^2$.

The volume of the solid is

$$V = \pi \int_0^4 [(4x)^2 - (x^2)^2] dx = \pi \int_0^4 (16x^2 - x^4) dx = \frac{2048\pi}{15}$$

14. $y = \sin x$, $y = \cos x$, $x = 0$, $y = 0$

The volume of the solid is

$$V = \pi \int_0^{\pi/4} (\cos^2 x - \sin^2 x) dx = \pi \int_0^{\pi/4} \cos 2x dx = \frac{\pi}{2}$$

15. $y = e^x$, $y = 0$, $x = 0$, $x = \ln 3$

The volume of the solid is

$$V = \pi \int_0^{\ln 3} e^{2x} dx = 4\pi$$

16. $y = e^{-2x}$, $y = 0$, $x = 0$, $x = 1$

The volume of the solid is

$$V = \pi \int_0^1 e^{-4x} dx = \frac{\pi}{4} (1 - e^{-4})$$

17. $y = \frac{1}{\sqrt{4+x^2}}$, $x = -2$, $x = 2$, $y = 0$

The volume of the solid is

$$\begin{aligned} V &= \pi \int_{-2}^2 \left(\frac{1}{\sqrt{4+x^2}} \right)^2 dx = 2\pi \int_0^2 \frac{1}{4+x^2} dx \\ &= 2\pi \times \frac{1}{2} \left[\tan^{-1} \frac{x}{2} \right]_0^2 = \frac{\pi^2}{4} \end{aligned}$$

18. $y = \frac{e^{3x}}{\sqrt{1+e^{6x}}}$, $x = 0$, $x = 1$, $y = 0$

The volume of the solid is

$$V = \pi \int_0^1 \frac{e^{6x}}{1+e^{6x}} dx = \frac{\pi}{6} [\ln(1+e^{6x})]_0^1 = \frac{\pi}{6} [\ln(1+e^6) - \ln 2]$$

Find the volume of the solid that results when the region enclosed by the given curves is revolved about the y-axis:

21. $x = \csc y, y = \pi/4, y = 3\pi/4, x = 0$

The volume of the solid is

$$V = \pi \int_{\pi/4}^{3\pi/4} \csc^2 y \, dy = \pi [-\cot y]_{\pi/4}^{3\pi/4} = \pi (\cot \pi/4 - \cot(3\pi/4)) = 2\pi$$

22. $y = x^2, x = y^2$

If the given graphs intersect, then

$$\sqrt{y} = y^2 \Rightarrow y = 0, 1$$

At $y = 0$, we have $x = 0$ and at $y = 1$, we have $x = 1$

Hence $(0,0)$ and $(1,1)$ are the points of intersection.

The volume of the solid is

$$V = \pi \int_0^1 (y - y^4) \, dy = \frac{3\pi}{10}$$

23. $x = y^2, x = y + 2$

If the given graphs intersect, then

$$y^2 = y + 2 \Rightarrow y = -1, 2$$

At $y = -1$, we have $x = 1$ and at $y = 2$, we have $x = 4$

Hence $(1,-1)$ and $(4,2)$ are the points of intersection.

The volume of the solid is

$$V = \pi \int_{-1}^2 [(y+2)^2 - y^4] \, dy = \frac{72\pi}{5}$$

24. $x = 1 - y^2, x = 2 + y^2, y = -1, y = 1$

The points of intersections are $(\frac{2}{3}, -1), (\frac{2}{3}, 1), (0,1)$ and $(0,-1)$.

The volume of the solid is

$$V = \pi \int_{-1}^1 [(2 + y^2)^2 - (1 - y^2)^2] \, dy = \pi \int_{-1}^1 (3 + 6y^2) \, dy = 10\pi$$

Note: For Figures, please see the class lectures.